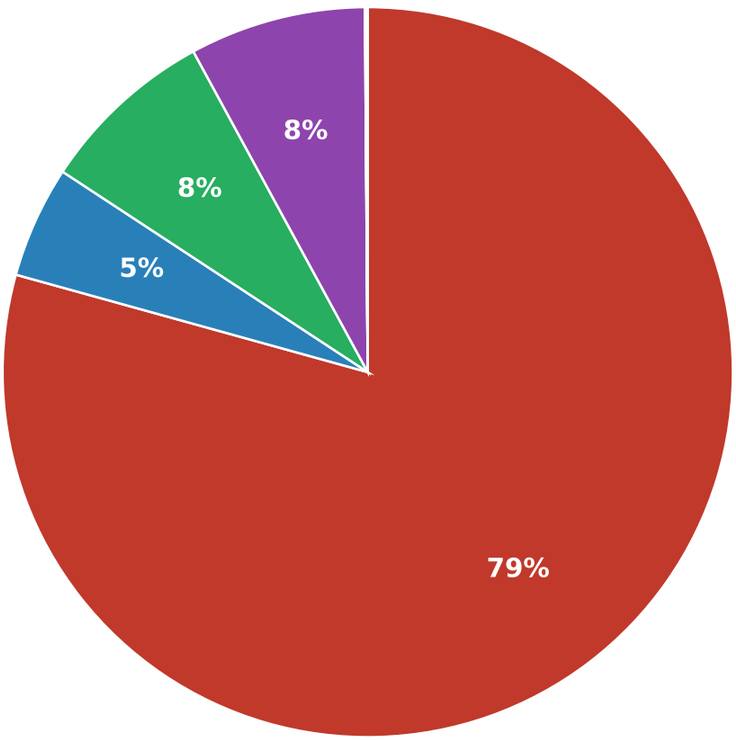
SpeechNet, one update step on GAP9 (GVSoC), board L1/L2 settings: what occupies L2 at the peak

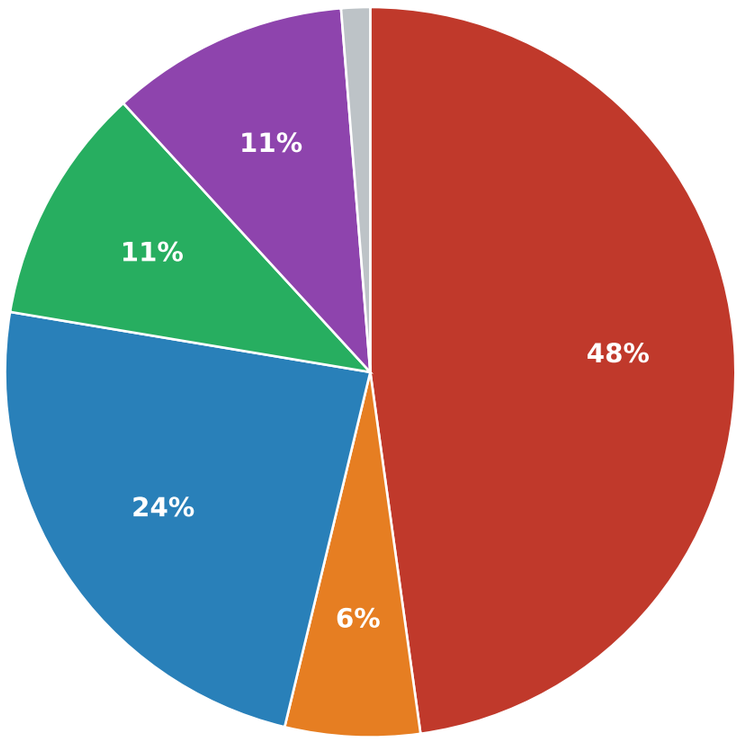
ZO (fp32)
L2 peak 792,032 B (schedule step 29)



- Block-0 activations: 628,096 B (79.3%)
- Data window + label: 39,208 B (5.0%)
- Trainable parameters (master): 61,956 B (7.8%)
- Perturbed parameter copies: 61,796 B (7.8%)
- BN auxiliary outputs (block 0): 128 B (0.0%)
- Static constants: 848 B (0.1%)

- Block-0 activations: Conv0 output, fp32 14x701x8, HWC (314,048 B); BN0 output, fp32 14x701x8, HWC (314,048 B)
- Data window + label: data window, fp32 14x700 (39,200 B); label (8 B)
- Trainable parameters (master): 22 tensors = the graph's trainable inputs (input_2..23): conv W and b, BN gamma and beta of blocks 0-4, FC W and b; live for the whole step (the update graph changes them in place)
- Perturbed parameter copies: 20 tensors = theta + eps*z of each trainable tensor (conv W stored transposed to HWC), all created at the start of the schedule; the 2 copies of block 0's conv W/b are already freed (block 0's conv has run)
- BN auxiliary outputs (block 0): 4 extra outputs written by block 0's BatchNormInternal (running mean/var out, saved mean/inv_std), unused, alive only around that node, 128 B
- Static constants: 11 compile-time constant tensors (multipliers, shifts)

QZO (int8, unfused)
L2 peak 164,248 B (schedule step 29)



- Block-0 activations: 78,512 B (47.8%)
- Conv input (HWC copy): 9,800 B (6.0%)
- Data window + label: 39,208 B (23.9%)
- Trainable parameters (master): 17,316 B (10.5%)
- Perturbed parameter copies: 17,316 B (10.5%)
- Static constants: 2,096 B (1.3%)

- Block-0 activations: Conv0 output, int8 14x701x8, HWC (78,512 B)
- Conv input (HWC copy): Conv0 input, int8 14x700, HWC copy (9,800 B)
- Data window + label: data window, fp32 14x700 (39,200 B); label (8 B)
- Trainable parameters (master): 22 tensors = the graph's trainable inputs (input_2..23): conv W and b, BN gamma and beta of blocks 0-4, FC W and b; live for the whole step (the update graph changes them in place)
- Perturbed parameter copies: 22 tensors = theta + eps*z of each trainable tensor (conv W stored transposed to HWC), all created at the start of the schedule
- Static constants: 26 compile-time constant tensors (multipliers, shifts)